

Sriaditya Vedantam

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EDUCATION

University of Georgia

Master of Science (M.S.) in Cybersecurity and Privacy

Emphasis in Cryptography & Privacy Preserving Machine Learning

Athens, GA, USA

Expected Spring 2028

University of Georgia

Bachelor of Science (B.S.) in Computer Science & Mathematics

Emphasis in Theoretical Computer Science, Cybersecurity, Pure Mathematics

Athens, GA, USA

Spring 2026

RESEARCH EXPERIENCE

Small Satellite Research Lab, University of Georgia

Athens, GA

Data Scientist Team Lead

August 2024 – May 2026

- Implementing and optimizing neural network applications with computational topology.
- Led a research engineering team of 12 members by defining short-term milestones increasing departmental output by over 80%.
- Working with NeRF (Neural Radiance Fields) with satellite imaging and drone data utilizing ML/CV along with implementation of perceptron with a more tensor based perspective.
- Utilized many ML frameworks, deciding on PyTorch.
- Later projects consisted of optimization of data transmission from space leading to viability of transmitters and fields of radii for wavelengths over a designated area.
- Area of transmission was less than 50 miles, leading to requirement for accuracy of data (image and video transmission), and scope.
- Determined viability for grants of over \$100,000 from NASA and the Air Force Association.

Mathematics Department, University of Georgia

Athens, GA

Research Experience

January 2024 – December 2025

- Focus on Exceptional Lie Algebras and their connections to Coxeter-Dynkin Diagrams.
- Read numerous papers linking with topics in Algebraic Geometry gaining foundational experience to reading research and papers, fostering a direction of interest.
- Focus on Birch and Swinnerton-Dyer Conjecture, Discrete Log problem, and elliptic curves applications in cryptography.
- Implemented early ECC algorithms with perspectives from math and looked into PQC Isogeny-Based Proposed Cryptographic Algorithms to understand fallacies in them, such as in SIKE.
- Focus on Kähler Manifolds and Complex Geometry.
- Focus on Elliptic Curves.
- Focus on Fermat's Little Theorem.

Cybersecurity Team, ImaginaryCTF

Leuven, Belgium

Board Member

November 2020 – May 2023

- Increased platform engagement by hosting an international competition with 60,000 participants.
- Coordinating operations and prize logistics with sponsors from Google, TryHackMe, OtterSec, and DigitalOcean.
- Improved challenge quality and integrity by designing cryptography problems grounded in current research.
- Utilized and implemented research papers from the International Association for Cryptologic Research (IACR).
- Oversaw teams to enhance efficiency and ensure distribution across network systems for projects and software.

Mathematics Department, Georgia Institute of Technology

Atlanta, GA

Research Project Developer

August 2021 – December 2021

- Delivered a graph theory application by implementing Dijkstra's algorithm with Eulerian circuit logic.
- Implemented the application in Java and integrated mapping API calls from Open Maps for real-world route use.

TEACHING EXPERIENCE

Mathematics Department, University of Georgia

Athens, GA

Peer Learning Assistant

January 2023 – December 2023

- Helped co-instruct as an assistant with a Lead Professor for Calculus II and III for Scientists and Engineers.
- Present curated material from Dr. Jacob Hicks often along with traversing the class and providing individual help and reasoning through problems.
- Provided lecture, review of the homework and classwork, facilitate discussion amongst students, and held office hours every week and by appointment.

SELECTED PUBLICATIONS / PRESENTATIONS

Vedantam, S., & Wang, H. (2023). Directed Reading Program Student Seminar. Athens, GA; Mathematics Department in the University of Georgia. Retrieved from <https://www.math.uga.edu/directed-reading-program>.

Vedantam, S., & Wang, H. (2023a). Directed Reading Program Student Seminar. Athens, GA; Mathematics Department in the University of Georgia. Retrieved from <https://www.math.uga.edu/directed-reading-program>.

OTHER EXPERIENCE

Athens Micro

Watkinsville, GA

Solutions Engineer

July 2026 – Present

- Build internal automation and API integrations across IT, dispatch, HR, and operations using PowerShell, TypeScript, SQL, n8n, and REST APIs
- Integrate platforms including ConnectWise Manage, NinjaOne, and Claude API to streamline workflows, ticketing, reporting, and internal processes.
- Scope projects independently, assess feasibility, set priorities and deadlines, troubleshoot issues, and deliver solutions directly to stakeholders and clients.

University of Georgia

Athens, GA

Parking Operations Representative

March 2024 – July 2026

- Engineered an automation pipeline using Selenium to execute 40,000+ system tasks, eliminating a 6-month operational backlog.
- Improved system reliability and reporting accuracy through repeatable workflow design.
- Resolved system bottlenecks with 60% efficiency increase in operational and reporting processes.
- Maintained compliance with financial and operational data handling standards.
- Applied secure design principles when implementing systems and automation workflows.

Cornerstone Entertainment Company

Raleigh, NC

Community Specialist Manager

January 2021 – December 2021

- Improved community retention and engagement by stabilizing staffing operations for a network of 200 users.
- Increased customer satisfaction 75% with coordination with developers to resolve platform reliability.

INVITED LECTURES

Crash Course on Competitive Cryptography (Lecture Series)

10/27/2026 - 10/29/2026

University of Georgia – The Hack Pack – Society for Cybersecurity

Athens, GA

Rational Solutions to Pythagorean Triples

11/29/2023

University of Georgia – Mathematics Student Seminar

Athens, GA

Primes and Fakes, Carmichael and the Twisted Prime Omega Function

04/27/2023

University of Georgia – Mathematics Student Seminar

Athens, GA

SKILLS

Fullstack Development, ML/CV Frameworks, Cloud Infrastructure, Technical Writing and Reading, Database Administration, C, C++, C#, Java, Python, SQL, SageMATH and JupyterLab

LANGUAGES

English – Native
Telugu – Native

SERVICE PROJECTS

Math Club , <i>VP of Communications and Promotion</i>	Fall 2025 – Present
The Hack Pack , <i>Red Team</i>	Fall 2023 – Present
Hats On Cybersecurity , <i>President (Merged into The Hack Pack)</i>	Fall 2022 – Fall 2023

CERTIFICATIONS

Security+ , <i>CompTIA</i>	May 2021 – May 2024
MTA Software Development Associate , <i>Microsoft</i>	April 2021
MTA Networking Associate , <i>Microsoft</i>	November 2020

HONORS AND AWARDS

Kossack Exam	2023
• Third Place Award out of 50 participants	
FBLA National Finalist for Management Information Systems	2022
• Placed 8 th place nationally single-handedly representing Georgia in a Case Study Competition	
1st Place Georgia, CyberPatriot XIV	2022
• Ended 1 st place in Georgia and Top 30 in the US as a Semifinalist for the Platinum Division in CyberPatriot.	
National Cyber Scholar	2021
• Place 51 st place out of over 3000 qualifying competitors and over 10,000 competitors.	
• Second Place Holder in Georgia.	
• Awarded \$3,000 Scholarship.	
FBLA National Finalist for Management Information Systems	2021
• Placed 7 th place nationally single-handedly representing Georgia in a Case Study Competition	
1st Place Georgia, CyberPatriot XIII	2021
• Ended 1 st place in Georgia and Top 30 in the US as a Semifinalist for the Platinum Division in CyberPatriot.	
1st Place Georgia, CyberPatriot XII	2020
• Ended 1 st place in Georgia and Top 30 in the US as a Semifinalist for the Platinum Division in CyberPatriot.	
Semifinalist, CyberPatriot XI	2019
• Ended as a Semifinalist and Top 100 in the US for the Gold Division in CyberPatriot.	
Semifinalist, CyberPatriot X	2018
• Ended as a Semifinalist and Top 100 in the US for the Middle School Division in CyberPatriot.	
22nd Place / 7794 - The Teamless A - picoCTF 2022	2022
• Cybersecurity Capture the Flag Competition	
23rd Place / 560 - TAMU Cyber Club - UTCTF 2022	2022
• Cybersecurity Capture the Flag Competition	
48th Place / 6481 - NCL Fall 2021 Individual	2021
• Cybersecurity Capture the Flag Competition	
35th Place / 3917 - Egg Heads - NCL Fall 2021 Team	2021
• Cybersecurity Capture the Flag Competition	
44th Place / 210 – The Teamless - pbctf 2021	2021
• Cybersecurity Capture the Flag Competition	
4th Place / 300 – ducks0ci3ty - DeconstruCT.F 2021	2021

• Cybersecurity Capture the Flag Competition	
37th Place / 1721 – ducks0ci3ty – H@cktivityCon 2021	2021
• Cybersecurity Capture the Flag Competition	
140th Place / 1216 – ducks0ci3ty - CSAW '21 Qualifiers	2021
• Cybersecurity Capture the Flag Competition	
140th Place / 4740 – ducks0ci3ty - HackTheBox Cyber Apocalypse 2021	2021
• Cybersecurity Capture the Flag Competition	
29th Place / 4180 - NCL Spring 2021 Individual	2021
• Cybersecurity Capture the Flag Competition	
22nd Place / 922 - Egg Heads - NCL Spring 2021 Team	2021
• Cybersecurity Capture the Flag Competition	
282nd Place / 1059 - ducks0ci3ty - DiceCTF 2021	2021
• Cybersecurity Capture the Flag Competition	
3rd Place / 12 - KSU ISA CTF 2020	2020
• Cybersecurity Capture the Flag Competition	
9th Place / 1064 - ducks0ci3ty - X-MAS CTF 2020	2020
• Cybersecurity Capture the Flag Competition	
38th Place / 6013 - NCL Fall 2020 Individual	2020
• Cybersecurity Capture the Flag Competition	
27th Place / 957 - Egg Heads - NCL Fall 2020 Team	2020
• Cybersecurity Capture the Flag Competition	
76th Place / 968 – Egg Heads - Newark Academy CTF 2020	2020
• Cybersecurity Capture the Flag Competition	
65th Place / 483 – Egg Heads (Solo) - TJCTF 2019	2019
• Cybersecurity Capture the Flag Competition	
71st Place / 1570 – Egg Heads - ångstromCTF 2019	2019
• Cybersecurity Capture the Flag Competition	
33rd Place / 60595 – Egg Heads (Solo) - PicoCTF 2019	2019
• Cybersecurity Capture the Flag Competition	

COURSEWORK

CSCI 1302: Software Development	Fall 2022
FYOS 1001: Effects of AI in the Modern Working World (Dr. Suchi Bhandarkar)	Fall 2022
MATH 2250: Calculus I for Scientists and Engineers	Fall 2022
CSCI 2610: Discrete Mathematics for Computer Science	Spring 2023
MATH 2260: Calculus II for Scientists and Engineers	Spring 2023
MATH 3200: Introduction to Mathematical Proofs	Spring 2023
MATH 2700: Elementary Differential Equations	Summer 2023
MATH 2500: Accelerated Calculus III for Engineers	Fall 2023
MATH 3100: Introduction to Mathematical Analysis (prev. Sequences and Series)	Fall 2023
MATH 6000: Abstract Algebra I (prev. Modern Algebra and Geometry I)	Fall 2023
CSCI 1730: Systems Programming	Fall 2023
MATH 3000: Introduction to Linear Algebra	Spring 2024
MATH 6010: Abstract Algebra II (prev. Modern Algebra and Geometry II)	Spring 2024
MATH 4950: Research in Mathematics	Spring 2024
CSCI 2720: Data Structures	Summer 2024
MATH 8300: Introduction to Algebraic Geometry	Fall 2024
CSCI 2670: Introduction to Theory of Computing	Spring 2025
CSCI 4370: Database Management	Spring 2025
MATH 8330: Hodge Theory (Audit)	Spring 2025
MATH 8200: Algebraic Topology (Audit)	Spring 2025

CSCI 4720: Computer Architecture and Organization	Summer 2025
CSCI 4760: Computer Networks	Summer 2025
CSCI 4380: Data Mining	Fall 2025
CSCI 4670: Combinatorics	Fall 2025
MATH 6100: Real Analysis	Fall 2025
MATH 8000: Algebra (Audit)	Fall 2025
CSCI 4470: Algorithms	Spring 2026
CSCI 4800: Human-Computer Interaction	Spring 2026
CSCI 4300: Web Programming	Spring 2026
CSCI 6730: Operating Systems	Fall 2026
CSCI 6760: Computer Networks	Fall 2026
CSCI 6950: Graduate Research	Fall 2026
CSCI 8000: Special Topics in Applied Cryptography	Fall 2026
CSCI 8960: Privacy Preserving Data Analysis / Machine Learning (Differential Privacy)	Fall 2026